Lecture 7: Continuity

Mathematical Analysis

1 Continuity

Definition 1.1 (Function Continuity at a Point)**.** [def:function-continuity] Under metric space (A, d_A) and (B, d_B), let $f : A \to B$. f is continous at a point $c \in A$ if

$$\forall \varepsilon : \mathbb{R}_+, \exists \delta : \mathbb{R}_+ \, \forall x \in B_\delta(c), \, d_B(f(x), f(c)) < \varepsilon$$

Lemma 1.1. *If a function f's limit exists at a point $c \in A$ which is $\ell \in B$, and f is continuous at c and $\ell = f(c)$*

Lemma 1.2. *Every isolated point c of a set A is a continous point*

Proposition 1.1. *A point c is continous on function f if and only if either its limit exists and equals to $f(c)$ or it is an isolated point of the domain*

2 Properties of Continuous Points

3 Continous Functions on Compact Sets

4 Extreme Value Theorem

5 Uniform Continuity

6 Intermediate Value Theorem